

PCB Design Process Output Matrix

PCB process stage map

PCB Step	Generated Output	Package Folder	Current Status
1. Schematic Creation / Validation	Native KiCad schematic, schematic PDF/SVG, dual-NCE/TFLN overview, visibility index.	04_front_end_schematic	Available. Parity closure still required.
2. BOM	Current BOM workbook, grouped/expanded CSV, schematic BOM, BOM action items, final BOM release template.	05_bom	Release pending until ECO/parity closure and component signoff.
3. Footprint Creation / Validation	Footprint-symbol reconciliation matrix and schematic parity issue detail.	06_footprints_symbols	84 footprint-symbol mismatches and 82 field mismatches remain open.
4. Netlist	KiCad XML netlist and S-expression netlist, plus netlist release summary.	07_netlists	Available. Reissue final after parity closure.
5. Board Outline / Mechanical Detail	295.0 x 140.0 mm mechanical envelope, 56.0 mm height note, KiCad PCB, STEP if available.	08_board_outline_mechanical	Envelope modeled. Real chassis signoff pending.
6. Stackup / Constraints Setting	Stackup/constraints requirements and manufacturer confirmation checklist.	09_stackup_constraints	Review checklist issued. Manufacturer stackup confirmation required.
7. Component Placement	Placement ECO instructions and current position CSV if available.	10_component_placement	Release pending due to edge-clearance and parity fixes.
8. Routing and Planes	Routing/planes review instructions and candidate layer preview.	11_routing_planes	Release pending due to no-net copper edge-clearance DRC items.
9. Silkscreen / DRC / FAB Notes	Silkscreen/fab-note review plus DRC fix list.	12_silkscreen_drc_fab_notes	4 silkscreen clipping items remain open.
10. Gerber Generation	Review Gerbers, drill, drill map/report where available, Gerber index.	13_gerbers_drill	Review outputs included. Regenerate final after DRC/parity closure.
11. DFA	Assembly/DFA review checklist and action matrix.	14_dfa_assembly	Checklist issued. Assembly release depends on final BOM.
12. DFM	DFM/manufacturability checklist and action matrix.	15_dfm_manufacturability	Checklist issued. DFM closure depends on ECO-clean outputs.